BAENGNYEONGDO, Korea, Republic of

WMO: **471020**

Lat: **37.974N** Lon: **124.713E** Elev: **38** StdP: **100.87** Time Zone: **9.00 (E09)** Period: **01-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR 99.6%			Humidification DP/MCDB and HR 99%			Coldest Month WS/MCDB 0.4%		Coldest Month WS/MCDB 1%		MCWS/PCWD to 99.6% DB		WSF	
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	**1**	**-9.1**	**-7.1**	**-16.7**	**0.9**	**-6.0**	**-15.3**	**1.0**	**-5.5**	**20.2**	**-1.8**	**15.6**	**-1.1**	**5.4**	**290**	**0.519**	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB 0.4%		Cooling DB/MCWB 1%		Cooling DB/MCWB 2%		Evaporation WB/MCDB 0.4%		Evaporation WB/MCDB 1%		Evaporation WB/MCDB 2%		MCWS/PCWD to 0.4% DB		
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	**8**	**4.2**	**28.3**	**24.5**	**27.1**	**23.7**	**26.0**	**23.0**	**25.6**	**27.1**	**24.9**	**26.3**	**24.1**	**25.5**	**3.5**	**230**	(2)

	Dehumidification DP/MCDB and HR 0.4%			Dehumidification DP/MCDB and HR 1%			Dehumidification DP/MCDB and HR 2%			Enthalpy/MCDB 0.4%		Enthalpy/MCDB 1%		Enthalpy/MCDB 2%		Extreme Max WB	
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	**25.2**	**20.4**	**26.6**	**24.4**	**19.5**	**25.8**	**23.6**	**18.5**	**25.2**	**79.4**	**27.2**	**76.2**	**26.4**	**73.1**	**25.5**	**29.8**	(3)

Extreme Annual Design Conditions

	Extreme Annual WS 1%	2.5%	5%		Extreme Annual Temperature Mean Min	Mean Max	Standard Deviation Min	Standard Deviation Max	n=5 years Min	n=5 years Max	n=10 years Min	n=10 years Max	n=20 years Min	n=20 years Max	n=50 years Min	n=50 years Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	**14.6**	**11.8**	**10.1**	DB	**-11.1**	**30.4**	**2.9**	**1.2**	**-13.2**	**31.3**	**-14.8**	**32.0**	**-16.4**	**32.7**	**-18.5**	**33.6**	(4)
(5)				WB	**-12.1**	**26.2**	**2.5**	**1.3**	**-13.9**	**27.2**	**-15.4**	**27.9**	**-16.8**	**28.7**	**-18.7**	**29.6**	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	**11.4**	**-1.3**	**0.1**	**3.9**	**9.2**	**14.7**	**19.2**	**22.5**	**23.9**	**20.4**	**14.9**	**7.8**	**1.2**	(6)
(7)			DBStd	**9.19**	**3.46**	**3.47**	**2.91**	**2.90**	**2.74**	**2.20**	**2.01**	**2.09**	**1.97**	**2.86**	**3.80**	**3.82**	(7)
(8)			HDD10.0	**1230**	**351**	**278**	**190**	**48**	**1**	**0**	**0**	**0**	**0**	**4**	**86**	**273**	(8)
(9)			HDD18.3	**2937**	**609**	**512**	**448**	**274**	**116**	**17**	**1**	**0**	**6**	**108**	**316**	**531**	(9)
(10)			CDD10.0	**1754**	**0**	**0**	**1**	**24**	**147**	**276**	**387**	**432**	**311**	**157**	**21**	**0**	(10)
(11)			CDD18.3	**418**	**0**	**0**	**0**	**0**	**4**	**42**	**129**	**173**	**67**	**3**	**0**	**0**	(11)
(12)			CDH23.3	**1402**	**0**	**0**	**0**	**0**	**4**	**48**	**389**	**889**	**71**	**1**	**0**	**0**	(12)
(13)			CDH26.7	**142**	**0**	**0**	**0**	**0**	**0**	**1**	**32**	**109**	**1**	**0**	**0**	**0**	(13)
(14)	Wind		WSAvg	**4.7**	**4.8**	**4.8**	**5.3**	**5.8**	**5.4**	**4.4**	**4.3**	**3.9**	**3.5**	**4.2**	**5.2**	**5.2**	(14)
(15)	Precipitation		PrecAvg	**818**	**13**	**18**	**19**	**50**	**72**	**76**	**219**	**166**	**86**	**35**	**40**	**22**	(15)
(16)			PrecMax	**1265**	**28**	**66**	**72**	**99**	**137**	**202**	**400**	**327**	**362**	**94**	**154**	**44**	(16)
(17)			PrecMin	**406**	**2**	**0**	**2**	**6**	**3**	**16**	**74**	**36**	**6**	**1**	**10**	**7**	(17)
(18)			PrecStd	**201**	**7**	**16**	**18**	**28**	**44**	**45**	**93**	**87**	**105**	**27**	**37**	**12**	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		**6.8**	**9.2**	**12.9**	**18.9**	**23.8**	**26.1**	**29.3**	**30.2**	**26.4**	**22.2**	**16.8**	**10.7**	(19)
(20)			MCWB		**4.9**	**6.0**	**6.4**	**10.7**	**14.1**	**18.5**	**25.0**	**25.8**	**20.9**	**17.5**	**13.9**	**8.9**	(20)
(21)		2%	DB		**5.7**	**6.8**	**10.4**	**16.7**	**21.8**	**24.7**	**27.4**	**28.9**	**25.1**	**20.6**	**15.3**	**9.2**	(21)
(22)			MCWB		**3.6**	**4.4**	**6.6**	**9.9**	**13.3**	**18.4**	**23.9**	**25.1**	**20.7**	**16.7**	**12.7**	**7.2**	(22)
(23)		5%	DB		**4.5**	**5.5**	**8.9**	**15.0**	**20.4**	**23.5**	**26.2**	**27.9**	**24.1**	**19.4**	**14.1**	**7.7**	(23)
(24)			MCWB		**2.2**	**3.3**	**5.6**	**9.1**	**13.5**	**18.3**	**23.4**	**24.4**	**20.0**	**15.9**	**11.8**	**5.2**	(24)
(25)		10%	DB		**3.4**	**4.4**	**7.7**	**13.6**	**19.1**	**22.5**	**25.3**	**26.8**	**23.1**	**18.4**	**13.0**	**6.5**	(25)
(26)			MCWB		**1.1**	**2.3**	**4.9**	**8.4**	**13.0**	**18.1**	**23.0**	**23.8**	**19.4**	**15.2**	**10.6**	**3.9**	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		**5.5**	**7.5**	**9.2**	**13.5**	**17.2**	**22.2**	**26.2**	**26.9**	**23.6**	**19.2**	**15.2**	**9.6**	(27)
(28)			MCDB		**6.5**	**8.4**	**10.2**	**16.3**	**20.2**	**23.9**	**27.2**	**29.0**	**24.7**	**20.7**	**16.1**	**10.4**	(28)
(29)		2%	WB		**4.0**	**5.0**	**7.7**	**11.7**	**16.1**	**20.9**	**25.1**	**26.1**	**22.1**	**18.0**	**13.7**	**7.5**	(29)
(30)			MCDB		**5.4**	**6.2**	**9.3**	**14.2**	**19.0**	**22.4**	**26.3**	**27.7**	**23.4**	**19.7**	**14.6**	**9.0**	(30)
(31)		5%	WB		**2.6**	**3.8**	**6.4**	**10.6**	**15.3**	**20.1**	**24.4**	**25.4**	**21.3**	**16.8**	**12.2**	**5.7**	(31)
(32)			MCDB		**4.2**	**5.2**	**8.0**	**12.9**	**18.1**	**21.8**	**25.6**	**26.9**	**22.8**	**18.6**	**13.5**	**7.5**	(32)
(33)		10%	WB		**1.2**	**2.6**	**5.4**	**9.6**	**14.4**	**19.5**	**23.7**	**24.7**	**20.5**	**15.9**	**11.0**	**4.2**	(33)
(34)			MCDB		**3.2**	**4.1**	**7.1**	**12.1**	**17.3**	**21.2**	**24.8**	**26.2**	**22.4**	**17.8**	**13.0**	**6.3**	(34)
(35)	Mean Daily Temperature Range		MDBR		**3.9**	**4.2**	**4.7**	**5.4**	**5.9**	**5.1**	**4.1**	**4.2**	**4.4**	**4.4**	**4.1**	**4.1**	(35)
(36)		5% DB	MCDBR		**4.8**	**5.3**	**6.5**	**7.4**	**7.7**	**6.5**	**5.4**	**5.1**	**5.3**	**5.2**	**4.3**	**4.9**	(36)
(37)			MCWBR		**4.7**	**4.6**	**4.5**	**4.4**	**4.2**	**3.7**	**3.1**	**3.0**	**3.5**	**3.8**	**4.1**	**4.9**	(37)
(38)		5% WB	MCDBR		**4.8**	**5.1**	**5.9**	**6.2**	**6.5**	**4.8**	**4.2**	**4.4**	**4.3**	**5.0**	**4.2**	**5.1**	(38)
(39)			MCWBR		**4.8**	**5.0**	**5.1**	**4.7**	**4.2**	**3.2**	**3.0**	**3.0**	**3.7**	**4.4**	**4.7**	**5.4**	(39)
(40)	Clear-Sky Solar Irradiance	taub			**0.439**	**0.494**	**0.554**	**0.605**	**0.624**	**0.706**	**0.682**	**0.585**	**0.515**	**0.526**	**0.507**	**0.446**	(40)
(41)		taud			**1.944**	**1.841**	**1.740**	**1.682**	**1.696**	**1.607**	**1.706**	**1.920**	**2.017**	**1.866**	**1.862**	**1.932**	(41)
(42)		Ebn at Noon			**688**	**703**	**705**	**700**	**698**	**642**	**653**	**709**	**736**	**664**	**610**	**642**	(42)
(43)		Edh at Noon			**138**	**174**	**212**	**238**	**239**	**261**	**235**	**185**	**159**	**168**	**150**	**131**	(43)
(44)	All-Sky Solar Radiation	RadAvg			**2.21**	**3.15**	**4.38**	**5.05**	**5.55**	**5.11**	**4.37**	**4.70**	**4.46**	**3.57**	**2.36**	**1.92**	(44)
(45)		RadStd			**0.17**	**0.23**	**0.36**	**0.35**	**0.51**	**0.62**	**0.59**	**0.57**	**0.38**	**0.25**	**0.27**	**0.13**	(45)

Historical Trends

		DBAvg	Heating 99% DB	Heating 99% DP	Cooling 1% DB	Cooling 1% WB	Cooling 1% DP	Degree-Days HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**+1.44**	**N/A**	**N/A**	**N/A**	**N/A**	(46)
(47)	Regional (16 neighbors)	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**+84**	**+48**	(47)

Nomenclature: See separate page